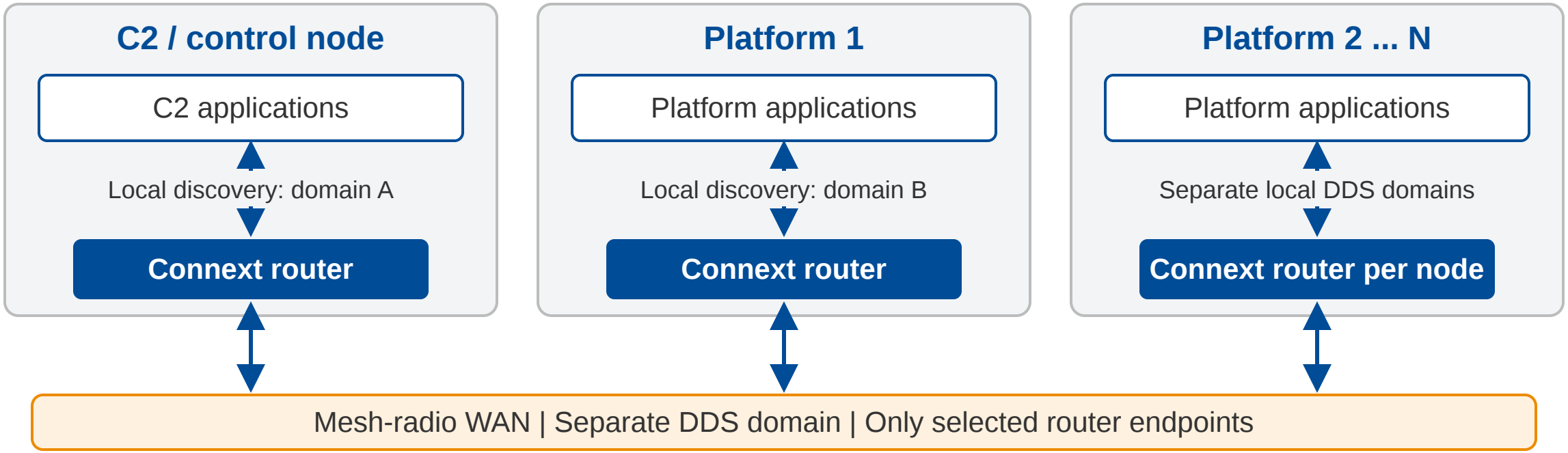
1. Isolate local discovery with a router



Start here: do not carry every local application's discovery over the radio.

2. Reduce what the WAN announces

Lever	What it reduces
Fewer WAN participants and endpoints	The number of discovery announcements
Disable type propagation for Connext < 7.7	Type definitions carried in discovery
Fewer shared WAN topics	Endpoint discovery, when they replace many per-topic endpoints

Before disabling types: provide compatible type definitions at every WAN consumer.
Applications and tools that learn types dynamically still need type propagation.

> **Separate the levers:** content filters reduce selected user data;
> they do not remove the endpoints from discovery.

3. Choose multicast or unicast discovery

	Multicast	Unicast
How	Announce to a shared group	Contact configured peer addresses
Gain	No per-platform address roster	Works without multicast
Cost	Radio must support efficient, reliable multicast	Maintain contacts; announcements may be sent to multiple peers
Best fit	Mesh with proven multicast behavior	Mesh where multicast is unavailable or costly

Discovery multicast does not require user-data multicast.

> **Measure on the radio:** compare airtime and discovery/recovery time.
> Neither option is automatically cheaper on a multi-hop mesh.

4. If unicast, choose how peers get addresses

Strategy	Simple model	Main trade-off
Configured peers	Load known peer addresses	Simple baseline; update when addresses change
C2 directory	Mission control supplies peer addresses	Application owns directory updates and recovery
Cloud Discovery Service (CDS)	RTI service introduces participants	Operate a reachable rendezvous service

C2 and CDS help peers find each other; they are not user-data relays.
Knowing one ordinary DDS peer does not automatically reveal its peers.

> **Practical sequence:** isolate -> reduce announcements -> choose transport -> choose peer setup.
> Validate with the same fleet size, loss, and restart scenarios.